

A CENTRAL LIMIT THEOREM FOR RANDOM EULER-PRODUCT SURROGATES OF $\log |\zeta(1/2 + it)|$

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ABSTRACT. We study the independent-phase Euler-product surrogate for $\log |\zeta(1/2 + it)|$ arising in classical zeta-function and probabilistic number theory [10, 1, 2, 12]. For independent ξ_p uniform on the unit circle, set

$$Z_y(t) := \prod_{p \leq y} (1 - \xi_p p^{-1/2 - it})^{-1}, \quad X_y(t) := \Re \log Z_y(t).$$

The decomposition $X_y(t) = S_y(t) + R_y(t)$ combines Lindeberg–Feller for the bounded prime layer [3, 4, 7, 8, 6], Fourier orthogonality for the uniformly L^2 -bounded remainder [3, 5], and Slutsky’s theorem [3, 4, 6]. Mertens’ theorem [1, 2, 11] gives

$$\sigma_y^2 := \frac{1}{2} \sum_{p \leq y} \frac{1}{p} \sim \frac{1}{2} \log \log y, \quad \frac{X_y(t)}{\sigma_y} \Rightarrow \mathcal{N}(0, 1).$$

The law is independent of t , and the numerical experiments record both finite-cutoff Gaussianity and the slow growth of σ_y^2 .

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Part 1. Random Euler products and Gaussian theory

1. INTRODUCTION

The Euler product suggests replacing the arithmetic phases p^{-it} by independent rotations [10, 12, 1, 2]. Throughout, p is prime, $y \geq 2$, $\Rightarrow$ denotes convergence in distribution, and

$$\theta_p \sim \text{Unif}([0, 2\pi)), \quad \xi_p := e^{i\theta_p}, \quad \mathbb{E}(\xi_p^k) = 0 \quad (k \in \mathbb{Z} \setminus \{0\}), \quad (\theta_p)_p \text{ independent.}$$

Thus $(\xi_p)_p$ is a Steinhaus family [12, 2]. The model retains the Euler weights and isolates the independent-prime mechanism behind Gaussian limits; it is not an identity in law with $\zeta(1/2 + it)$ [3, 5, 10, 1, 7, 2].

1.1. Random Euler product.

Definition 1.1.1 (Random Euler product, [10, 12, 1]). For $t \in \mathbb{R}$, define

$$Z_y(t) := \prod_{p \leq y} (1 - \xi_p p^{-1/2-it})^{-1}, \quad \log Z_y(t) := \sum_{p \leq y} \sum_{k \geq 1} \frac{\xi_p^k p^{-ikt}}{k p^{k/2}}, \quad X_y(t) := \Re \log Z_y(t).$$

$$|\xi_p p^{-1/2-it}| < 1 \xRightarrow{2.1.1} \log Z_y(t) = \sum_{p \leq y} -\log(1 - \xi_p p^{-1/2-it}).$$

Definition 1.1.2 (Layers and variance scale, [5, 2]). Set

$$S_y(t) := \sum_{p \leq y} \frac{\cos(\theta_p - t \log p)}{\sqrt{p}}, \quad R_y(t) := \sum_{p \leq y} \sum_{k \geq 2} \frac{\cos(k(\theta_p - t \log p))}{k p^{k/2}}, \quad \sigma_y^2 := \frac{1}{2} \sum_{p \leq y} \frac{1}{p}.$$

Let $B_1 = 0.2614972 \dots$ be the Meissel–Mertens constant and set

$$\Phi(x) := \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-u^2/2} du, \quad \mathbf{Y}_y(t) := (X_y(t), S_y(t), R_y(t)).$$

The proof mechanism is

$$y \rightarrow \infty \xRightarrow{2.3.5, 2.2.7} \left(\frac{S_y}{\sigma_y} \Rightarrow \mathcal{N}(0, 1) \wedge \frac{R_y}{\sigma_y} \xrightarrow{\mathbb{P}} 0 \right) \xRightarrow{3.1.5} \frac{S_y + R_y}{\sigma_y} \Rightarrow \mathcal{N}(0, 1) \xLeftrightarrow{2.1.2} \frac{X_y}{\sigma_y} \Rightarrow \mathcal{N}(0, 1).$$

1.2. Results.

Theorem A. *The random Euler product satisfies the following.*

(1) For every $m \in \mathbb{N}$, $h, t_1, \dots, t_m \in \mathbb{R}$, and $y \geq 2$,

$$\mathcal{L}\left((\mathbf{Y}_y(t_j + h))_{j=1}^m\right) = \mathcal{L}\left((\mathbf{Y}_y(t_j))_{j=1}^m\right).$$

Proof. § 4. □

(2) For every $t \in \mathbb{R}$, with $C_R := \frac{1}{2} \sum_p \sum_{k \geq 2} (k^2 p^k)^{-1}$, one has

$$\sigma_y^2 = \frac{1}{2} \log \log y + \frac{B_1}{2} + O\left(\frac{1}{\log y}\right), \quad \text{Var}(X_y(t)) = \frac{1}{2} \log \log y + \frac{B_1}{2} + C_R + O\left(\frac{1}{\log y}\right).$$

Proof. § 4. □

(3) For every $t \in \mathbb{R}$,

$$\frac{X_y(t)}{\sigma_y} \Rightarrow \mathcal{N}(0, 1) \quad (y \rightarrow \infty).$$

Proof. § 4. □

Theorem B. *Let $C_{\text{BE}} > 0$ be the constant in Theorem 3.1.3. The following hold uniformly in $t \in \mathbb{R}$.*

(1) For every $y \geq 2$,

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P} \left(\frac{S_y(t)}{\sigma_y} \leq x \right) - \Phi(x) \right| \leq \frac{4C_{\text{BE}}}{3\pi\sigma_y^3} \sum_{p \leq y} p^{-3/2} < \frac{C_{\text{BE}}}{2\sigma_y^3}.$$

Proof. § 4. □

(2) If $\sigma_y \geq 1$, then

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P} \left(\frac{X_y(t)}{\sigma_y} \leq x \right) - \Phi(x) \right| \leq \left(\frac{1}{\sqrt{2\pi}} + \frac{1}{6} + \frac{C_{\text{BE}}}{2} \right) \sigma_y^{-2/3}.$$

Proof. § 4. □

Part 1 continues with § 2, § 3, and the proofs of Theorems A–B in § 4; Part 2 contains the supplementary tools and code.

2. PRIME LAYERS AND GAUSSIAN APPROXIMATION

Definitions 1.1.1 and 1.1.2 reduce the model to the logarithmic expansion, circle orthogonality, and prime harmonic sums [3, 5, 10, 1, 11, 2]. Put

$$U_p(t) := (\theta_p - t \log p) \bmod 2\pi, \quad z_p := \xi_p p^{-1/2 - it}.$$

2.1. Expansion and phase symmetry.

Lemma 2.1.1 (Absolutely convergent logarithm). *For $p \leq y$,*

$$|z_p| = p^{-1/2} \leq 2^{-1/2} < 1, \quad \sum_{k \geq 1} \frac{|z_p|^k}{k} \leq \frac{p^{-1/2}}{1 - p^{-1/2}} \leq 1 + \sqrt{2},$$

and

$$\log Z_y(t) = \sum_{p \leq y} \sum_{k \geq 1} \frac{e^{ik(\theta_p - t \log p)}}{kp^{k/2}} = \sum_{p \leq y} -\log(1 - z_p).$$

Proof. Use the power series of $-\log(1 - z)$.

$$|z_p| = p^{-1/2} \leq 2^{-1/2} < 1 \implies -\log(1 - z_p) = \sum_{k \geq 1} \frac{z_p^k}{k}.$$

$$\sum_{k \geq 1} \frac{|z_p|^k}{k} \leq \sum_{k \geq 1} p^{-k/2} = \frac{p^{-1/2}}{1 - p^{-1/2}} \leq 1 + \sqrt{2}.$$

$$\#\{p : p \leq y\} < \infty \implies \sum_{p \leq y} \sum_{k \geq 1} \frac{|z_p|^k}{k} < \infty \xrightarrow{1.1.1} \log Z_y(t) = \sum_{p \leq y} -\log(1 - z_p).$$

This proves absolute convergence and the expansion. □

Corollary 2.1.2 (Layer decomposition). *For $y \geq 2$ and $t \in \mathbb{R}$,*

$$X_y(t) = \sum_{p \leq y} \sum_{k \geq 1} \frac{\cos(k(\theta_p - t \log p))}{kp^{k/2}} = S_y(t) + R_y(t).$$

Proof. Take real parts in Lemma 2.1.1.

$$z_p^k = p^{-k/2} e^{ik(\theta_p - t \log p)} \implies \Re(z_p^k) = p^{-k/2} \cos(k(\theta_p - t \log p)).$$

$$X_y(t) = \Re \log Z_y(t) \xrightarrow{1.1.2, 1.1} X_y(t) = \sum_{p \leq y} \sum_{k \geq 1} \frac{\cos(k(\theta_p - t \log p))}{kp^{k/2}}.$$

$$\sum_{k \geq 1} = \sum_{k=1} + \sum_{k \geq 2} \xrightarrow{1.1.2} X_y(t) = S_y(t) + R_y(t).$$

This proves the decomposition. □

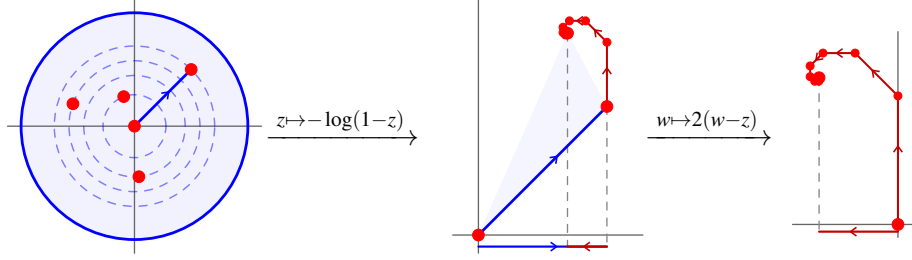


FIGURE 1. Sampled prime radii lead to $z = (1+i)/2$, whose logarithmic spiral splits into a blue linear term and red higher-power tail; real projection gives $X_y(t) = S_y(t) + R_y(t)$.

Lemma 2.1.3 (Uniform shift symmetry). *If $\Theta \sim \text{Unif}([0, 2\pi))$ and $\alpha \in \mathbb{R}$, then*

$$(\Theta - \alpha) \bmod 2\pi \sim \text{Unif}([0, 2\pi)).$$

Proof. Use translation invariance of circular Lebesgue measure.

$$\tau_\alpha(u) := (u - \alpha) \bmod 2\pi, \quad |\tau_\alpha^{-1}(I)| = |I|.$$

$$\mathbb{P}((\Theta - \alpha) \bmod 2\pi \in I) = \mathbb{P}(\Theta \in \tau_\alpha^{-1}(I)) = \frac{|\tau_\alpha^{-1}(I)|}{2\pi} = \frac{|I|}{2\pi}.$$

$$\mathbb{P}(\tau_\alpha(\Theta) \in I) = \frac{|I|}{2\pi} \iff \tau_\alpha(\Theta) \sim \text{Unif}([0, 2\pi)).$$

This proves shift invariance. □

Lemma 2.1.4 (Circle orthogonality). *If $U \sim \text{Unif}([0, 2\pi))$ and $k, \ell \geq 1$, then*

$$\mathbb{E}(e^{ikU}) = 0, \quad \mathbb{E}[\cos(kU)] = \mathbb{E}[\sin(kU)] = 0,$$

$$\mathbb{E}[\cos(kU) \cos(\ell U)] = \mathbb{E}[\sin(kU) \sin(\ell U)] = \frac{1}{2} \delta_{k\ell},$$

$$\mathbb{E}[\cos(kU) \sin(\ell U)] = 0.$$

Proof. Use Fourier orthogonality on the circle.

$$\mathbb{E}(e^{imU}) = \frac{1}{2\pi} \int_0^{2\pi} e^{imu} du = \delta_{m0} \quad (m \in \mathbb{Z}).$$

$$2\cos(kU) \cos(\ell U) = \cos((k-\ell)U) + \cos((k+\ell)U),$$

$$2\sin(kU) \sin(\ell U) = \cos((k-\ell)U) - \cos((k+\ell)U),$$

$$\implies \mathbb{E}[\cos(kU) \cos(\ell U)] = \mathbb{E}[\sin(kU) \sin(\ell U)] = \frac{1}{2} \delta_{k\ell}$$

$$\cos(kU) \sin(\ell U) = \frac{1}{2} (\sin((k+\ell)U) - \sin((k-\ell)U)) \implies \mathbb{E}[\cos(kU) \sin(\ell U)] = 0.$$

This proves all identities. □

Proposition 2.1.5 (The one-point law is independent of t). *For every $y \geq 2$ and $t \in \mathbb{R}$, the variables $(U_p(t))_{p \leq y}$ are i.i.d. uniform, and*

$$\mathcal{L}(X_y(t)) = \mathcal{L}(X_y(0)), \quad \mathcal{L}(S_y(t)) = \mathcal{L}(S_y(0)), \quad \mathcal{L}(R_y(t)) = \mathcal{L}(R_y(0)).$$

Proof. Apply Lemma 2.1.3 primewise.

$$\begin{aligned} U_p(t) &= (\theta_p - t \log p) \bmod 2\pi \xrightarrow{2.1.3} U_p(t) \sim \text{Unif}([0, 2\pi)). \\ (\theta_p)_{p \leq y} \text{ independent} \wedge U_p(t) &= \tau_{t \log p}(\theta_p) \implies (U_p(t))_{p \leq y} \text{ independent.} \\ (U_p(t))_{p \leq y} &\stackrel{d}{=} (U_p(0))_{p \leq y} \xrightarrow{1.1.1, 1.1.2, 2.1.2} \mathbf{Y}_y(t) \stackrel{d}{=} \mathbf{Y}_y(0). \end{aligned}$$

This proves the three law identities. $\square$

Proposition 2.1.6 (Stationarity and covariance, [12, 2]). *For $m \in \mathbb{N}$, $h, t_1, \dots, t_m \in \mathbb{R}$, and $y \geq 2$,*

$$\mathcal{L}\left((\mathbf{Y}_y(t_j + h))_{j=1}^m\right) = \mathcal{L}\left((\mathbf{Y}_y(t_j))_{j=1}^m\right).$$

Moreover,

$$\begin{aligned} \text{Cov}(S_y(t), S_y(s)) &= \frac{1}{2} \sum_{p \leq y} \frac{\cos((t-s) \log p)}{p}, \\ \text{Cov}(X_y(t), X_y(s)) &= \frac{1}{2} \sum_{p \leq y} \sum_{k \geq 1} \frac{\cos(k(t-s) \log p)}{k^2 p^k}. \end{aligned}$$

Proof. Use one common rotation and circle orthogonality.

$$\begin{aligned} \theta'_p &:= (\theta_p - h \log p) \bmod 2\pi \xrightarrow{2.1.3} (\theta'_p)_{p \leq y} \stackrel{d}{=} (\theta_p)_{p \leq y} \implies (U_p(t_j + h))_{p,j} \stackrel{d}{=} (U_p(t_j))_{p,j}. \\ \mathbb{E}[\cos(\theta_p - a) \cos(\theta_p - b)] &= \frac{1}{2} \cos(a - b) \xrightarrow{2.1.4} \text{Cov}(S_y(t), S_y(s)) = \frac{1}{2} \sum_{p \leq y} \frac{\cos((t-s) \log p)}{p}. \\ \mathbb{E}[\cos(k(\theta_p - a)) \cos(\ell(\theta_p - b))] &= \frac{1}{2} \delta_{k\ell} \cos(k(a - b)). \end{aligned}$$

$$X_y(u) \stackrel{2.1.2}{=} \sum_{p \leq y} \sum_{k \geq 1} \frac{\cos(k U_p(u))}{k p^{k/2}} \quad (u \in \{s, t\}) \implies \text{Cov}(X_y(t), X_y(s)) = \frac{1}{2} \sum_{p \leq y} \sum_{k \geq 1} \frac{\cos(k(t-s) \log p)}{k^2 p^k}.$$

This proves stationarity and both covariance formulas. $\square$

2.2. Moments and remainder control.

Proposition 2.2.1 (Moments and isotropy of the main layer). *Set*

$$V_y(t) := \sum_{p \leq y} \frac{e^{i(\theta_p - t \log p)}}{\sqrt{p}}, \quad S_y(t) = \Re V_y(t), \quad T_y(t) := \Im V_y(t).$$

Then

$$\mathbb{E}[V_y(t)] = 0, \quad \mathbb{E}|V_y(t)|^2 = \sum_{p \leq y} \frac{1}{p} = 2\sigma_y^2, \quad \mathbb{E}[S_y(t)] = \mathbb{E}[T_y(t)] = 0,$$

and

$$\begin{bmatrix} \text{Var}(S_y(t)) & \text{Cov}(S_y(t), T_y(t)) \\ \text{Cov}(T_y(t), S_y(t)) & \text{Var}(T_y(t)) \end{bmatrix} = \sigma_y^2 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Proof. Independence and circle orthogonality determine all entries.

$$U_p(t) \sim \text{Unif}([0, 2\pi)) \xrightarrow{2.1.4} (\mathbb{E}e^{iU_p(t)}, \mathbb{E}\cos^2 U_p(t), \mathbb{E}\sin^2 U_p(t), \mathbb{E}[\cos U_p(t) \sin U_p(t)]) = \left(0, \frac{1}{2}, \frac{1}{2}, 0\right),$$

$$\implies \mathbb{E}[S_y(t)] = \mathbb{E}[T_y(t)] = 0.$$

$$\mathbb{E}[V_y(t)] = \sum_{p \leq y} \frac{\mathbb{E}(e^{iU_p(t)})}{\sqrt{p}} = 0, \quad \mathbb{E}|V_y(t)|^2 = \sum_{p, q \leq y} \frac{\mathbb{E}(e^{i(U_p(t) - U_q(t))})}{\sqrt{pq}} = \sum_{p \leq y} \frac{1}{p} = 2\sigma_y^2.$$

$$\text{Var}(S_y(t)) = \text{Var}(T_y(t)) = \frac{1}{2} \sum_{p \leq y} \frac{1}{p} = \sigma_y^2, \quad \text{Cov}(S_y(t), T_y(t)) = \sum_{p \leq y} \frac{\mathbb{E}[\cos U_p(t) \sin U_p(t)]}{p} = 0.$$

This proves the moment and covariance formulas. $\square$

Theorem 2.2.2 (Mertens, [1, 11, 2]). As $y \rightarrow \infty$,

$$\sum_{p \leq y} \frac{1}{p} = \log \log y + B_1 + O\left(\frac{1}{\log y}\right), \quad \sigma_y^2 = \frac{1}{2} \log \log y + \frac{B_1}{2} + O\left(\frac{1}{\log y}\right), \quad \sigma_y \rightarrow \infty.$$

Proposition 2.2.3 (First product moments, [1, 11, 2]). For every $y \geq 2$ and $t \in \mathbb{R}$,

$$\mathbb{E}Z_y(t) = 1, \quad \mathbb{E}|Z_y(t)|^2 = \prod_{p \leq y} \left(1 - \frac{1}{p}\right)^{-1} \sim e^\gamma \log y,$$

where γ is Euler's constant.

Proof. Circle orthogonality evaluates each Euler factor.

$$U \sim \text{Unif}([0, 2\pi)) \wedge 0 < r < 1 \implies \mathbb{E}\left(\frac{1}{1 - re^{iU}}\right) = \sum_{m \geq 0} r^m \mathbb{E}(e^{imU}) = 1.$$

$$\mathbb{E}\left(\frac{1}{|1 - re^{iU}|^2}\right) = \sum_{m, n \geq 0} r^{m+n} \mathbb{E}(e^{i(m-n)U}) = \sum_{m \geq 0} r^{2m} = \frac{1}{1 - r^2}.$$

$$r = p^{-1/2} \wedge (U_p(t))_{p \leq y} \text{ independent} \implies \mathbb{E}Z_y(t) = 1, \quad \mathbb{E}|Z_y(t)|^2 = \prod_{p \leq y} \left(1 - \frac{1}{p}\right)^{-1} \sim e^\gamma \log y.$$

This proves both product moments. $\square$

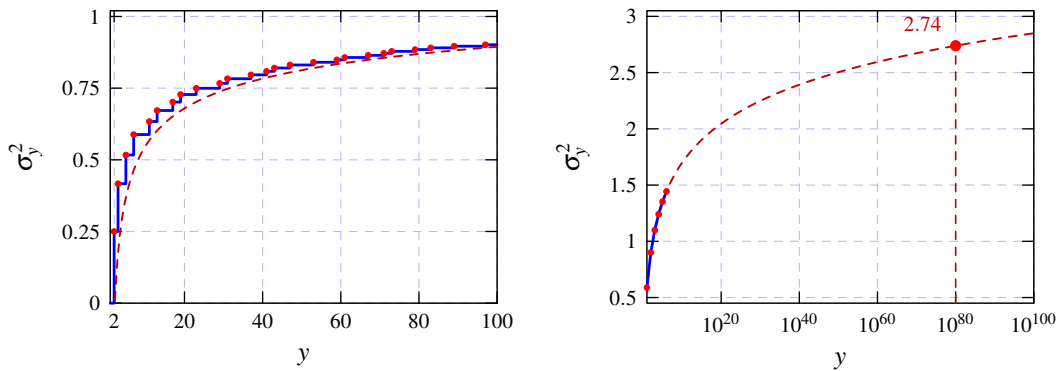


FIGURE 2. Exact small-cutoff variance follows the Mertens curve, illustrating its exceptionally slow double-logarithmic growth.

Remark 2.2.4 (Cutoff scale, [1, 11, 2]). For $y_a := \inf\{y \geq 2 : \sigma_y^2 \geq a\}$, Theorem 2.2.2 gives

$$\log \log y_a = 2a - B_1 + o(1), \quad \log_{10} y_a = \frac{e^{2a - B_1 + o(1)}}{\log 10}, \quad \sigma_y^2 = 10 \implies y \approx 10^{1.62 \cdot 10^8}.$$

Lemma 2.2.5 (One-prime remainder). *For*

$$R_p(t) := \sum_{k \geq 2} \frac{\cos(kU_p(t))}{kp^{k/2}},$$

one has

$$|R_p(t)| \leq \frac{p^{-1}}{1 - p^{-1/2}} \leq \frac{2 + \sqrt{2}}{p}, \quad |R_y(t)| \leq 2(2 + \sqrt{2})\sigma_y^2, \\ \mathbb{E}[R_p(t)] = 0, \quad \text{Var}(R_p(t)) = \frac{1}{2} \sum_{k \geq 2} \frac{1}{k^2 p^k} \leq \frac{1}{4p^2}.$$

Proof. Absolute convergence and circle orthogonality control one prime.

$$|R_p(t)| \leq \sum_{k \geq 2} p^{-k/2} = \frac{p^{-1}}{1 - p^{-1/2}} \leq \frac{2 + \sqrt{2}}{p} \implies |R_y(t)| \leq (2 + \sqrt{2}) \sum_{p \leq y} \frac{1}{p} = 2(2 + \sqrt{2})\sigma_y^2. \\ U_p(t) \sim \text{Unif}([0, 2\pi)) \xrightarrow{2.1.4} \mathbb{E}[R_p(t)] = 0, \quad \text{Var}(R_p(t)) = \frac{1}{2} \sum_{k \geq 2} \frac{1}{k^2 p^k}. \\ \frac{1}{2} \sum_{k \geq 2} \frac{1}{k^2 p^k} \leq \frac{1}{8} \sum_{k \geq 2} p^{-k} = \frac{1}{8p(p-1)} \leq \frac{1}{4p^2}.$$

This proves the pointwise and variance bounds. □

Proposition 2.2.6 (Mean-square bounded remainder). *For every $y \geq 2$ and $t \in \mathbb{R}$,*

$$\mathbb{E}[R_y(t)] = 0, \quad \text{Var}(R_y(t)) = \frac{1}{2} \sum_{p \leq y} \sum_{k \geq 2} \frac{1}{k^2 p^k} \leq \frac{1}{4}(\zeta(2) - 1) < \frac{1}{6}, \\ \text{Cov}(S_y(t), R_y(t)) = 0, \quad \text{Var}(X_y(t)) = \sigma_y^2 + \text{Var}(R_y(t)).$$

Proof. Primewise independence and orthogonality split the two layers.

$$\mathbb{E}[R_y(t)] = \sum_{p \leq y} \mathbb{E}[R_p(t)] = 0, \quad \text{Var}(R_y(t)) = \sum_{p \leq y} \text{Var}(R_p(t)) \xrightarrow{2.2.5} \text{Var}(R_y(t)) = \frac{1}{2} \sum_{p \leq y} \sum_{k \geq 2} \frac{1}{k^2 p^k}. \\ \text{Var}(R_y(t)) \leq \frac{1}{4} \sum_{p \leq y} \frac{1}{p^2} < \frac{1}{4} \sum_{n \geq 2} \frac{1}{n^2} = \frac{1}{4}(\zeta(2) - 1) < \frac{1}{6}. \\ \text{Cov}(S_y(t), R_y(t)) = \sum_{p \leq y} \sum_{k \geq 2} \frac{\mathbb{E}[\cos U_p(t) \cos(kU_p(t))]}{kp^{(k+1)/2}} \xrightarrow{2.1.4} \text{Cov}(S_y(t), R_y(t)) = 0, \\ X_y \stackrel{2.1.2}{=} S_y + R_y \wedge \text{Var}(S_y) \stackrel{2.2.1}{=} \sigma_y^2 \wedge \text{Cov}(S_y, R_y) = 0 \implies \text{Var}(X_y(t)) = \sigma_y^2 + \text{Var}(R_y(t)).$$

This proves the mean-square decomposition. □

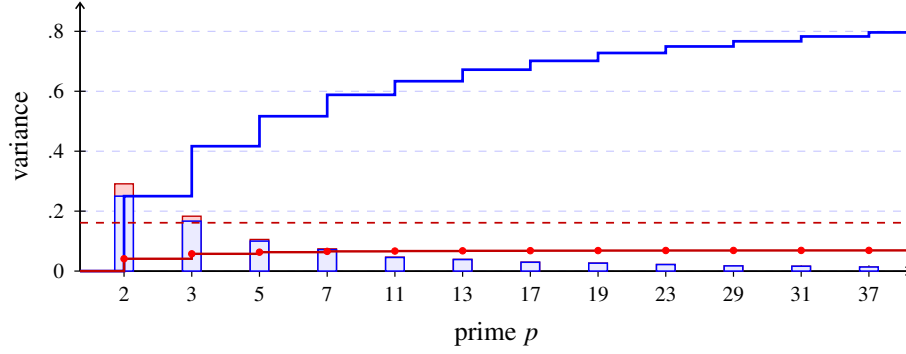


FIGURE 3. Primewise variances split into the divergent blue scale σ_y^2 and a bounded red higher-power remainder.

Corollary 2.2.7 (Negligible remainder). *Uniformly in $t \in \mathbb{R}$,*

$$\mathbb{E} \left[\left(\frac{R_y(t)}{\sigma_y} \right)^2 \right] < \frac{1}{6\sigma_y^2} \rightarrow 0, \quad \frac{R_y(t)}{\sigma_y} \xrightarrow{\mathbb{P}} 0, \quad \frac{X_y(t) - S_y(t)}{\sigma_y} \xrightarrow{\mathbb{P}} 0.$$

Proof. The L^2 bound and Chebyshev force the normalized tail to vanish.

$$0 \leq \mathbb{E} \left[\left(\frac{R_y(t)}{\sigma_y} \right)^2 \right] = \frac{\text{Var}(R_y(t))}{\sigma_y^2} < \frac{1}{6\sigma_y^2} \xrightarrow{2.2.6, 2.2.2} \mathbb{E} \left[\left(\frac{R_y(t)}{\sigma_y} \right)^2 \right] \rightarrow 0.$$

$$\varepsilon > 0 \implies \sup_{t \in \mathbb{R}} \mathbb{P}(|R_y(t)| > \varepsilon \sigma_y) \leq \frac{1}{\varepsilon^2} \sup_{t \in \mathbb{R}} \mathbb{E} \left[\left(\frac{R_y(t)}{\sigma_y} \right)^2 \right] < \frac{1}{6\varepsilon^2 \sigma_y^2} \rightarrow 0.$$

$$\sup_{t \in \mathbb{R}} \mathbb{P} \left(\frac{|R_y(t)|}{\sigma_y} > \varepsilon \right) \rightarrow 0 \implies \frac{R_y(t)}{\sigma_y} \xrightarrow{\mathbb{P}} 0 \xLeftrightarrow{2.1.2} \frac{X_y(t) - S_y(t)}{\sigma_y} \xrightarrow{\mathbb{P}} 0.$$

This proves all three uniform limits. $\square$

2.3. Gaussian approximation. The main layer is a centered triangular array of independent bounded summands whose variance diverges. We use Lindeberg–Feller [3, 8, 4, 6, 7] and then record a characteristic-function proof [3, 5, 6].

Definition 2.3.1 (Prime-indexed triangular array, [3, 4]). For $p \leq y$, set

$$X_{p,y}(t) := \frac{\cos U_p(t)}{\sqrt{p}}, \quad S_y(t) = \sum_{p \leq y} X_{p,y}(t), \quad |X_{p,y}(t)| \leq p^{-1/2} \leq 2^{-1/2}.$$

Lemma 2.3.2 (Lindeberg condition, [3, 8, 4, 6, 7]). *Uniformly in $t \in \mathbb{R}$,*

$$\mathbb{E}[X_{p,y}(t)] = 0, \quad \sum_{p \leq y} \text{Var}(X_{p,y}(t)) = \sigma_y^2, \quad \max_{p \leq y} \frac{|X_{p,y}(t)|}{\sigma_y} \leq \frac{1}{\sqrt{2} \sigma_y} \rightarrow 0.$$

For every $\varepsilon > 0$,

$$\frac{1}{\sigma_y^2} \sum_{p \leq y} \mathbb{E} [X_{p,y}(t)^2 \mathbf{1}\{|X_{p,y}(t)| > \varepsilon \sigma_y\}] = 0$$

once $\sigma_y \geq (\varepsilon \sqrt{2})^{-1}$.

Proof. Use Lemma 2.1.4 and Theorem 2.2.2.

$$U_p(t) \sim \text{Unif}([0, 2\pi)) \xrightarrow{2.1.4} \mathbb{E}[X_{p,y}(t)] = 0, \quad \text{Var}(X_{p,y}(t)) = \frac{1}{2p}, \quad \sum_{p \leq y} \text{Var}(X_{p,y}(t)) = \sigma_y^2.$$

$$\sigma_y \rightarrow \infty \xrightarrow{2.2.2} \max_{p \leq y} \frac{|X_{p,y}(t)|}{\sigma_y} \leq \frac{1}{\sqrt{2}} \rightarrow 0.$$

$$\varepsilon > 0, \quad \sigma_y \geq (\varepsilon\sqrt{2})^{-1} \implies |X_{p,y}(t)| \leq 2^{-1/2} \leq \varepsilon\sigma_y \implies \mathbf{1}\{|X_{p,y}(t)| > \varepsilon\sigma_y\} = 0,$$

$$\mathbf{1}\{|X_{p,y}(t)| > \varepsilon\sigma_y\} = 0 \implies \frac{1}{\sigma_y^2} \sum_{p \leq y} \mathbb{E}[X_{p,y}(t)^2 \mathbf{1}\{|X_{p,y}(t)| > \varepsilon\sigma_y\}] = 0.$$

This proves the Lindeberg condition. $\square$

Theorem 2.3.3 (Lindeberg–Feller, [3, 8, 4, 6, 7]). *Let $\{X_{n,k} : 1 \leq k \leq k_n\}$ be independent within each row and satisfy*

$$\mathbb{E}[X_{n,k}] = 0, \quad s_n^2 := \sum_{k=1}^{k_n} \text{Var}(X_{n,k}) \in (0, \infty),$$

$$\frac{1}{s_n^2} \sum_{k=1}^{k_n} \mathbb{E}[X_{n,k}^2 \mathbf{1}\{|X_{n,k}| > \varepsilon s_n\}] \rightarrow 0 \quad (\varepsilon > 0).$$

Then

$$\frac{1}{s_n} \sum_{k=1}^{k_n} X_{n,k} \Rightarrow \mathcal{N}(0, 1).$$

Fact 2.3.4 (Cramér–Wold, [3, 4]). *For random vectors $Y_n, Y \in \mathbb{R}^d$,*

$$Y_n \Rightarrow Y \iff a \cdot Y_n \Rightarrow a \cdot Y \quad \text{for every } a \in \mathbb{R}^d.$$

Theorem 2.3.5 (Central limit theorem for the main layer, [3, 8, 4, 6, 7]). *For every $t \in \mathbb{R}$,*

$$\frac{S_y(t)}{\sigma_y} \Rightarrow \mathcal{N}(0, 1), \quad \mathcal{L}\left(\frac{S_y(t)}{\sigma_y}\right) = \mathcal{L}\left(\frac{S_y(0)}{\sigma_y}\right).$$

Proof. Apply Theorem 2.3.3 to the array in Definition 2.3.1.

$$\{X_{p,y}(t) : p \leq y\} \xrightarrow{2.3.1, 2.3.2} \text{independent and centered}, \quad \sum_{p \leq y} \text{Var}(X_{p,y}(t)) = \sigma_y^2.$$

$$\text{Lindeberg} \xrightarrow{2.3.2, 2.3.3} \frac{S_y(t)}{\sigma_y} = \frac{1}{\sigma_y} \sum_{p \leq y} X_{p,y}(t) \Rightarrow \mathcal{N}(0, 1).$$

$$(U_p(t))_{p \leq y} \stackrel{d}{=} (U_p(0))_{p \leq y} \xrightarrow{2.1.5} \mathcal{L}\left(\frac{S_y(t)}{\sigma_y}\right) = \mathcal{L}\left(\frac{S_y(0)}{\sigma_y}\right).$$

This proves both claims. $\square$

Corollary 2.3.6 (Isotropic Gaussian limit, [3, 4]). *For every $t \in \mathbb{R}$,*

$$\frac{(S_y(t), T_y(t))}{\sigma_y} \Rightarrow \mathcal{N}(0, I_2).$$

Proof. Use the Cramér–Wold device [3, 4].

$$\begin{aligned} (a, b) \in \mathbb{R}^2 \setminus \{(0, 0)\}, \quad \mathbf{a} := (a, b), \quad r := \sqrt{a^2 + b^2}, \quad (a, b) = r(\cos \alpha, \sin \alpha), \\ \implies \frac{aS_y(t) + bT_y(t)}{\sigma_y} = r \sum_{p \leq y} \frac{\cos(U_p(t) - \alpha)}{\sigma_y \sqrt{p}} \stackrel{d}{=} r \frac{S_y(t)}{\sigma_y}. \end{aligned}$$

$$\begin{aligned} \frac{S_y(t)}{\sigma_y} &\Rightarrow \mathcal{N}(0, 1) \xrightarrow{2.3.5} r \frac{S_y(t)}{\sigma_y} \Rightarrow \mathcal{N}(0, a^2 + b^2). \\ \mathbf{a} \cdot \frac{(S_y(t), T_y(t))}{\sigma_y} &\Rightarrow \mathcal{N}(0, \|\mathbf{a}\|_2^2) \quad (\mathbf{a} \in \mathbb{R}^2) \xLeftrightarrow{2.3.4} \frac{(S_y(t), T_y(t))}{\sigma_y} \Rightarrow \mathcal{N}(0, I_2). \end{aligned}$$

This proves the isotropic limit. $\square$

Corollary 2.3.7 (Asymptotic independence at distinct times, [3, 4, 1, 2]). *For $m \in \mathbb{N}$ and distinct $t_1, \dots, t_m \in \mathbb{R}$,*

$$\frac{(S_y(t_1), \dots, S_y(t_m))}{\sigma_y} \Rightarrow \mathcal{N}(0, I_m), \quad \frac{(X_y(t_1), \dots, X_y(t_m))}{\sigma_y} \Rightarrow \mathcal{N}(0, I_m).$$

Proof. Use prime-harmonic cancellation, Cramér–Wold, and Slutsky [3, 4, 1, 2].

$$j \neq k \implies \tau_{jk} := t_j - t_k \neq 0 \implies \sum_{p \leq y} \frac{e^{-i\tau_{jk} \log p}}{p} = O_{\mathbf{t}}(1) \xrightarrow{2.1.6} \text{Cov}(S_y(t_j), S_y(t_k)) = O_{\mathbf{t}}(1),$$

$$j = k \xrightarrow{2.2.1} \text{Cov}(S_y(t_j), S_y(t_k)) = \sigma_y^2 \implies \text{Cov}(S_y(t_j), S_y(t_k)) = \sigma_y^2 \delta_{jk} + O_{\mathbf{t}}(1).$$

$$\mathbf{a} = (a_1, \dots, a_m) \in \mathbb{R}^m \setminus \{0\}, \quad Y_{p,y} := \frac{1}{\sqrt{p}} \sum_{j=1}^m a_j \cos U_p(t_j),$$

$$v_y^2 := \sum_{p \leq y} \text{Var}(Y_{p,y}) = \sigma_y^2 \|\mathbf{a}\|_2^2 + O_{\mathbf{a}, \mathbf{t}}(1) \implies \frac{v_y}{\sigma_y} \longrightarrow \|\mathbf{a}\|_2.$$

$$|Y_{p,y}| \leq \frac{\|\mathbf{a}\|_1}{\sqrt{2}} = o(v_y) \implies \frac{1}{v_y^2} \sum_{p \leq y} \mathbb{E} [Y_{p,y}^2 \mathbf{1}\{|Y_{p,y}| > \varepsilon v_y\}] = 0,$$

$$\text{Lindeberg} \xrightarrow{2.3.3} \frac{\sum_{p \leq y} Y_{p,y}}{v_y} \Rightarrow \mathcal{N}(0, 1),$$

$$\frac{\sum_{j=1}^m a_j S_y(t_j)}{\sigma_y} = \frac{v_y}{\sigma_y} \frac{\sum_{p \leq y} Y_{p,y}}{v_y} \Rightarrow \mathcal{N}(0, \|\mathbf{a}\|_2^2) \quad (\mathbf{a} \in \mathbb{R}^m) \xLeftrightarrow{2.3.4} \frac{(S_y(t_1), \dots, S_y(t_m))}{\sigma_y} \Rightarrow \mathcal{N}(0, I_m).$$

$$\mathbf{X}_y := (X_y(t_j))_{j=1}^m, \quad \mathbf{S}_y := (S_y(t_j))_{j=1}^m, \quad \mathbf{R}_y := (R_y(t_j))_{j=1}^m, \quad \mathbf{X}_y \stackrel{2.1.2}{=} \mathbf{S}_y + \mathbf{R}_y,$$

$$\mathbb{P} \left(\max_{1 \leq j \leq m} \frac{|R_y(t_j)|}{\sigma_y} > \varepsilon \right) \leq \sum_{j=1}^m \mathbb{P} \left(\frac{|R_y(t_j)|}{\sigma_y} > \varepsilon \right) \xrightarrow{2.2.7} \mathbb{P} \left(\frac{\|\mathbf{R}_y\|_\infty}{\sigma_y} > \varepsilon \right) \longrightarrow 0.$$

$$\frac{\mathbf{S}_y}{\sigma_y} \Rightarrow \mathcal{N}(0, I_m) \wedge \frac{\mathbf{R}_y}{\sigma_y} \xrightarrow{\mathbb{P}} 0 \xrightarrow{3.1.5} \frac{\mathbf{X}_y}{\sigma_y} = \frac{\mathbf{S}_y + \mathbf{R}_y}{\sigma_y} \Rightarrow \mathcal{N}(0, I_m).$$

This proves both vector limits. $\square$

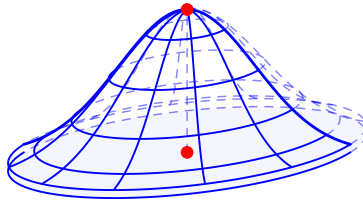


FIGURE 4. The normalized prime walk converges to an isotropic Gaussian, and its real projection gives the one-dimensional limit.

Lemma 2.3.8 (Bessel characteristic function, [3, 5]). *For $U \sim \text{Unif}([0, 2\pi])$ and $a \in \mathbb{R}$,*

$$\mathbb{E}(e^{ia \cos U}) = \frac{1}{2\pi} \int_0^{2\pi} e^{ia \cos u} du = \sum_{m \geq 0} \frac{(-1)^m}{(m!)^2} \left(\frac{a}{2}\right)^{2m} = J_0(a).$$

Moreover,

$$\begin{aligned} \log J_0(a) &= -\frac{a^2}{4} - \frac{a^4}{64} + O(a^6) \quad (a \rightarrow 0), \\ |a| \leq 1 &\implies J_0(a) \geq \frac{3}{4}, \quad \left| \log J_0(a) + \frac{a^2}{4} \right| \leq \frac{a^4}{24}. \end{aligned}$$

Proof. Expand the exponential and logarithm.

$$\cos^{2m} u = 4^{-m} \sum_{j=0}^{2m} \binom{2m}{j} e^{i(2m-2j)u} \xrightarrow{2.1.4} \mathbb{E}(\cos^{2m} U) = 4^{-m} \binom{2m}{m}, \quad \mathbb{E}(\cos^{2m+1} U) = 0.$$

$$\mathbb{E}(e^{ia \cos U}) = \sum_{m \geq 0} \frac{(ia)^{2m}}{(2m)!} 4^{-m} \binom{2m}{m} = \sum_{m \geq 0} \frac{(-1)^m}{(m!)^2} \left(\frac{a}{2}\right)^{2m} = J_0(a).$$

$$r(a) := \sum_{m \geq 2} \frac{(-1)^m}{(m!)^2} \left(\frac{a}{2}\right)^{2m}, \quad x := \frac{a^2}{4} - r(a) = 1 - J_0(a),$$

$$|a| \leq 1 \implies 0 \leq r(a) \leq \frac{a^4}{64} \wedge 0 \leq x \leq \frac{1}{4} \implies J_0(a) = 1 - x \geq \frac{3}{4}.$$

$$\log J_0(a) + \frac{a^2}{4} = r(a) - \sum_{n \geq 2} \frac{x^n}{n} \in \left[-\frac{x^2}{2(1-x)}, \frac{a^4}{64} \right] \subset \left[-\frac{a^4}{24}, \frac{a^4}{24} \right].$$

$$r(a) = \frac{a^4}{64} + O(a^6) \wedge x = \frac{a^2}{4} - \frac{a^4}{64} + O(a^6) \implies \log(1-x) = -\frac{a^2}{4} - \frac{a^4}{64} + O(a^6).$$

This proves the exact and local formulas. $\square$

Proposition 2.3.9 (Characteristic function, [3, 5, 6]). *For $\lambda, t \in \mathbb{R}$ and $\sigma_y \geq |\lambda|/\sqrt{2}$,*

$$\mathbb{E}(e^{i\lambda S_y(t)/\sigma_y}) = \prod_{p \leq y} J_0\left(\frac{\lambda}{\sigma_y \sqrt{p}}\right) > 0,$$

$$\left| \log \mathbb{E}(e^{i\lambda S_y(t)/\sigma_y}) + \frac{\lambda^2}{2} \right| \leq \frac{\lambda^4}{24\sigma_y^4} \sum_{p \leq y} \frac{1}{p^2} \leq \frac{\lambda^4}{48\sigma_y^4}, \quad \mathbb{E}(e^{i\lambda S_y(t)/\sigma_y}) \longrightarrow e^{-\lambda^2/2}.$$

Proof. Put $a_p := \lambda/(\sigma_y \sqrt{p})$.

$$\sigma_y \geq \frac{|\lambda|}{\sqrt{2}} \implies |a_p| \leq 1, \quad (U_p(t))_{p \leq y} \text{ i.i.d. uniform} \xrightarrow{2.1.5} \mathbb{E}(e^{i\lambda S_y(t)/\sigma_y}) = \prod_{p \leq y} \mathbb{E}(e^{ia_p \cos U_p(t)}).$$

$$\prod_{p \leq y} \mathbb{E}(e^{ia_p \cos U_p(t)}) \xrightarrow{2.3.8} \prod_{p \leq y} J_0(a_p) > 0, \quad \sigma_y^2 = \frac{1}{2} \sum_{p \leq y} \frac{1}{p} \xrightarrow{1.1.2} \sum_{p \leq y} \frac{a_p^2}{4} = \frac{\lambda^2}{2}.$$

$$\sum_p \frac{1}{p^2} \leq \frac{\pi^2}{8} - \frac{3}{4} < \frac{1}{2} \implies \left| \sum_{p \leq y} \log J_0(a_p) + \frac{\lambda^2}{2} \right| \xrightarrow{2.3.8} \sum_{p \leq y} \frac{a_p^4}{24} = \frac{\lambda^4}{24\sigma_y^4} \sum_{p \leq y} \frac{1}{p^2} \leq \frac{\lambda^4}{48\sigma_y^4}.$$

$$\sigma_y^2 = \frac{1}{2} \sum_{p \leq y} \frac{1}{p} \xrightarrow{2.2.2} \sigma_y \rightarrow \infty \implies \log \mathbb{E}(e^{i\lambda S_y(t)/\sigma_y}) \longrightarrow -\frac{\lambda^2}{2} \implies \mathbb{E}(e^{i\lambda S_y(t)/\sigma_y}) \longrightarrow e^{-\lambda^2/2}.$$

This proves the characteristic-function estimate. $\square$

Corollary 2.3.10 (First Gaussian correction, [8, 5, 6]). *For fixed $\lambda \in \mathbb{R}$, uniformly in t , as $y \rightarrow \infty$,*

$$\log \mathbb{E}(e^{i\lambda S_y(t)/\sigma_y}) = -\frac{\lambda^2}{2} - \frac{\lambda^4}{64\sigma_y^4} \sum_{p \leq y} \frac{1}{p^2} + O_\lambda(\sigma_y^{-6}).$$

Proof. Put $a_p := \lambda/(\sigma_y \sqrt{p})$.

$$\begin{aligned} \sigma_y^2 &= \frac{1}{2} \sum_{p \leq y} \frac{1}{p} \xrightarrow{2.2.2} \sigma_y \rightarrow \infty \implies \max_{p \leq y} |a_p| \rightarrow 0 \xrightarrow{2.3.8} \log J_0(a_p) = -\frac{a_p^2}{4} - \frac{a_p^4}{64} + O(a_p^6). \\ \sigma_y^2 &= \frac{1}{2} \sum_{p \leq y} \frac{1}{p} \xrightarrow{1.1.2} \sum_{p \leq y} \frac{a_p^2}{4} = \frac{\lambda^2}{2}, \quad \sum_{p \leq y} \frac{a_p^4}{64} = \frac{\lambda^4}{64\sigma_y^4} \sum_{p \leq y} \frac{1}{p^2}. \\ \sum_{p \leq y} a_p^6 &= \frac{\lambda^6}{\sigma_y^6} \sum_{p \leq y} \frac{1}{p^3} = O_\lambda(\sigma_y^{-6}) \implies \log \mathbb{E}(e^{i\lambda S_y(t)/\sigma_y}) = -\frac{\lambda^2}{2} - \frac{\lambda^4}{64\sigma_y^4} \sum_{p \leq y} \frac{1}{p^2} + O_\lambda(\sigma_y^{-6}). \end{aligned}$$

This proves the first correction. $\square$

3. RATES, COMPUTATION, AND ARITHMETIC SCOPE

The exact remainder variance and Berry–Esseen yield finite-cutoff rates [5, Chapter XVI], [7, Chapter 5]. Put

$$\varphi(x) := \frac{e^{-x^2/2}}{\sqrt{2\pi}}, \quad \text{Li}_2(x) := \sum_{k \geq 1} \frac{x^k}{k^2}, \quad P(s) := \sum_p p^{-s} \quad (s > 1),$$

and let C_{BE} be the absolute Berry–Esseen constant.

3.1. Remainder and rates.

Proposition 3.1.1 (Exact remainder variance). *For every $y \geq 2$ and $t \in \mathbb{R}$,*

$$\begin{aligned} \text{Var}(R_y(t)) &= \frac{1}{2} \sum_{p \leq y} (\text{Li}_2(p^{-1}) - p^{-1}) \uparrow C_R := \frac{1}{2} \sum_p (\text{Li}_2(p^{-1}) - p^{-1}) < \frac{1}{6}, \\ 0 \leq C_R - \text{Var}(R_y(t)) &\leq \frac{1}{4} \sum_{p > y} p^{-2} \leq \frac{1}{4(y-1)}, \quad \text{Var}(X_y(t)) = \sigma_y^2 + \text{Var}(R_y(t)). \end{aligned}$$

Proof. Expand the dilogarithm and use the layer decomposition.

$$\begin{aligned} \text{Li}_2(p^{-1}) - p^{-1} &= \sum_{k \geq 2} \frac{1}{k^2 p^k} \xrightarrow{2.2.6} \text{Var}(R_y(t)) = \frac{1}{2} \sum_{p \leq y} (\text{Li}_2(p^{-1}) - p^{-1}). \\ 0 < \frac{1}{2} \sum_{k \geq 2} \frac{1}{k^2 p^k} &\leq \frac{1}{8} \sum_{k \geq 2} p^{-k} = \frac{1}{8p(p-1)} \leq \frac{1}{4p^2} \implies 0 < C_R < \frac{1}{4} \sum_{n \geq 2} n^{-2} < \frac{1}{6}. \\ 0 \leq C_R - \text{Var}(R_y(t)) &\leq \frac{1}{4} \sum_{p > y} p^{-2} \leq \frac{1}{4} \sum_{n > [y]} n^{-2} \leq \frac{1}{4[y]} \leq \frac{1}{4(y-1)}. \end{aligned}$$

$$X_y(t) = S_y(t) + R_y(t) \wedge \text{Cov}(S_y(t), R_y(t)) = 0 \xrightarrow{2.1.2, 2.2.6} \text{Var}(X_y(t)) = \sigma_y^2 + \text{Var}(R_y(t)).$$

This proves all three formulas. $\square$

Corollary 3.1.2 (Limiting remainder). *For every $t \in \mathbb{R}$, there is $R_\infty(t) \in L^2$ with*

$$R_y(t) \longrightarrow R_\infty(t) \text{ in } L^2, \quad \mathbb{E}(R_\infty(t)) = 0, \quad \text{Var}(R_\infty(t)) = C_R.$$

Proof. Use completeness of L^2 [3, 4, 6].

$$v > y \implies \mathbb{E}(|R_v(t) - R_y(t)|^2) = \frac{1}{2} \sum_{y < p \leq v} \sum_{k \geq 2} \frac{1}{k^2 p^k} \leq \frac{1}{4} \sum_{p > y} p^{-2} \longrightarrow 0.$$

$$\mathbb{E}(|R_v(t) - R_y(t)|^2) \longrightarrow 0 \implies (R_y(t))_{y \geq 2} \text{ Cauchy in } L^2 \implies R_y(t) \longrightarrow R_\infty(t) \text{ in } L^2.$$

$$|\mathbb{E}(R_\infty(t))| \leq \mathbb{E}(|R_\infty(t) - R_y(t)|) \leq \|R_\infty(t) - R_y(t)\|_2 \longrightarrow 0 \implies \mathbb{E}(R_\infty(t)) = 0.$$

$$\|\|R_\infty(t)\|_2 - \|R_y(t)\|_2\| \leq \|R_\infty(t) - R_y(t)\|_2 \longrightarrow 0 \xrightarrow{3.1.1} \text{Var}(R_\infty(t)) = C_R.$$

This proves the limiting identities. $\square$

Theorem 3.1.3 (Berry–Esseen, [5, 7]). *Let $Y_1, \dots, Y_n$ be independent and centered with $\mathbb{E}(|Y_j|^3) < \infty$. Then*

$$s^2 := \sum_{j=1}^n \text{Var}(Y_j) > 0 \implies \sup_{x \in \mathbb{R}} \left| \mathbb{P} \left(s^{-1} \sum_{j=1}^n Y_j \leq x \right) - \Phi(x) \right| \leq \frac{C_{\text{BE}}}{s^3} \sum_{j=1}^n \mathbb{E}(|Y_j|^3).$$

Lemma 3.1.4 (Third absolute moments). *For every prime $p \leq y$,*

$$\mathbb{E}(|X_{p,y}(t)|^3) = \frac{4}{3\pi p^{3/2}}, \quad \sum_{p \leq y} \mathbb{E}(|X_{p,y}(t)|^3) < \frac{4P(3/2)}{3\pi} = 0.3605 \dots < \frac{1}{2}.$$

Proof. Circle integration gives the summand moments.

$$\theta_p \sim \text{Unif}([0, 2\pi)) \xrightarrow{2.1.3} U_p(t) \sim \text{Unif}([0, 2\pi)).$$

$$\mathbb{E}(|\cos(U_p(t))|^3) = \frac{2}{\pi} \int_0^{\pi/2} \cos^3(u) du = \frac{4}{3\pi}.$$

$$X_{p,y}(t) \xrightarrow{2.3.1} p^{-1/2} \cos(U_p(t)) \implies \mathbb{E}(|X_{p,y}(t)|^3) = \frac{4}{3\pi p^{3/2}}.$$

$$\sum_{p \leq y} \mathbb{E}(|X_{p,y}(t)|^3) < \frac{4}{3\pi} \sum_p p^{-3/2} = \frac{4P(3/2)}{3\pi} = 0.3605 \dots < \frac{1}{2}.$$

This proves both bounds. $\square$

Fact 3.1.5 (Chebyshev and Slutsky, [3, 4, 6]). *For $Y \in L^2$, $\varepsilon > 0$, and random variables V_n, W_n on one probability space,*

$$\mathbb{P}(|Y - \mathbb{E}(Y)| \geq \varepsilon) \leq \frac{\text{Var}(Y)}{\varepsilon^2}, \quad V_n \Rightarrow V \wedge W_n \xrightarrow{\mathbb{P}} 0 \implies V_n + W_n \Rightarrow V.$$

Lemma 3.1.6 (Normal comparison). *For $x \in \mathbb{R}$, $\delta > 0$, and $\varepsilon \in \{-1, 1\}$,*

$$|\Phi(x + \varepsilon\delta) - \Phi(x)| \leq \frac{\delta}{\sqrt{2\pi}}.$$

Proof. Integrate the standard normal density.

$$0 \leq \varphi(u) \leq \varphi(0) = \frac{1}{\sqrt{2\pi}} \implies \|\varphi\|_\infty = \frac{1}{\sqrt{2\pi}}.$$

$$\Phi(x + \varepsilon\delta) - \Phi(x) = \int_x^{x+\varepsilon\delta} \varphi(u) du.$$

$$|\Phi(x + \varepsilon\delta) - \Phi(x)| \leq \delta \|\varphi\|_\infty = \frac{\delta}{\sqrt{2\pi}}.$$

This proves the comparison. $\square$

Corollary 3.1.7 (Mertens-scale rates). *Uniformly in $t \in \mathbb{R}$, as $y \rightarrow \infty$,*

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P} \left(\frac{S_y(t)}{\sigma_y} \leq x \right) - \Phi(x) \right| = O \left((\log(\log(y)))^{-3/2} \right),$$

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P} \left(\frac{X_y(t)}{\sigma_y} \leq x \right) - \Phi(x) \right| = O \left((\log(\log(y)))^{-1/3} \right).$$

Proof. Use Theorem B and Mertens' estimate [1, 11, 2].

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P} \left(\frac{S_y(t)}{\sigma_y} \leq x \right) - \Phi(x) \right| \stackrel{\text{B}}{\Rightarrow} O(\sigma_y^{-3}).$$

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P} \left(\frac{X_y(t)}{\sigma_y} \leq x \right) - \Phi(x) \right| \stackrel{\text{B}}{\Rightarrow} O(\sigma_y^{-2/3}).$$

$$\sigma_y^2 \stackrel{2.2.2}{=} \frac{1}{2} \log(\log(y)) + O(1) \Rightarrow \sigma_y^{-3} = O \left((\log(\log(y)))^{-3/2} \right), \quad \sigma_y^{-2/3} = O \left((\log(\log(y)))^{-1/3} \right).$$

This proves both rates. $\square$

3.2. Finite-cutoff computation. The fourth moment is exact, while Listing 1 in Part 2 reproduces the sample.

Proposition 3.2.1 (Third and fourth moments). *For every $y \geq 2$ and $t \in \mathbb{R}$,*

$$\mathbb{E} \left(\left(\frac{S_y(t)}{\sigma_y} \right)^3 \right) = 0, \quad \mathbb{E} \left(\left(\frac{S_y(t)}{\sigma_y} \right)^4 \right) = 3 - \frac{3}{8\sigma_y^4} \sum_{p \leq y} p^{-2} < 3.$$

Proof. Use circle moments and the independent-sum formula [5, Chapter XV], [6, Chapter 4].

$$\cos^2(u) = \frac{1 + \cos(2u)}{2}, \quad \cos^3(u) = \frac{3 \cos(u) + \cos(3u)}{4}, \quad \cos^4(u) = \frac{3 + 4 \cos(2u) + \cos(4u)}{8}.$$

$$U \sim \text{Unif}([0, 2\pi)) \stackrel{2.1.4}{\Rightarrow} \mathbb{E}(\cos^2(U)) = \frac{1}{2}, \quad \mathbb{E}(\cos^3(U)) = 0, \quad \mathbb{E}(\cos^4(U)) = \frac{3}{8}.$$

$$\mathbb{E}(X_{p,y}(t)^2) = \frac{1}{2p}, \quad \mathbb{E}(X_{p,y}(t)^3) = 0, \quad \mathbb{E}(X_{p,y}(t)^4) = \frac{3}{8p^2}.$$

$$\mathbb{E}(S_y(t)^3) = \sum_{p \leq y} \mathbb{E}(X_{p,y}(t)^3) = 0, \quad \mathbb{E}(S_y(t)^4) = \sum_{p \leq y} \frac{3}{8p^2} + 6 \sum_{p < q \leq y} \frac{1}{4pq} = 3\sigma_y^4 - \frac{3}{8} \sum_{p \leq y} p^{-2}.$$

$$\text{Var}(S_y(t)) \stackrel{2.2.1}{=} \sigma_y^2 \Rightarrow \mathbb{E} \left(\left(\frac{S_y(t)}{\sigma_y} \right)^4 \right) = 3 - \frac{3}{8\sigma_y^4} \sum_{p \leq y} p^{-2} < 3.$$

This proves the moment formulas. $\square$

Definition 3.2.2 (Simulation protocol). Generate primes by the sieve of Eratosthenes [9, Chapter 1], reset `numpy.random.default_rng(7)` at each $y \in \{10^3, 10^4, 10^5\}$, and take $t = 0$, $N := 2 \cdot 10^4$,

$$\theta_p^{(j)} \text{ i.i.d. } \text{Unif}([0, 2\pi)), \quad Z_j := \frac{1}{\sigma_y} \sum_{p \leq y} \frac{\cos(\theta_p^{(j)})}{\sqrt{p}} \quad (1 \leq j \leq N).$$

$$\bar{Z} := \frac{1}{N} \sum_{j=1}^N Z_j, \quad \hat{v} := \frac{1}{N} \sum_{j=1}^N (Z_j - \bar{Z})^2, \quad \hat{\gamma}_r := \frac{1}{N} \sum_{j=1}^N \left(\frac{Z_j - \bar{Z}}{\sqrt{\hat{v}}} \right)^r \quad (r = 3, 4).$$

$$u_k := 0.0025 + \frac{0.995(k-1)}{98}, \quad j_k := 1 + \left\lfloor u_k(N-1) + \frac{1}{2} \right\rfloor, \quad q_k := \Phi^{-1} \left(\frac{j_k - \frac{1}{2}}{N} \right) \quad (1 \leq k \leq 99),$$

with $Z_{(1)} \leq \dots \leq Z_{(N)}$, quantile pairs $(q_k, Z_{(j_k)})$, and a 60-bin histogram on the sample range.

Proposition 3.2.3 (Consistency of the experiment). *For fixed y , let F_y be the distribution function of $S_y(t)/\sigma_y$ and*

$$\widehat{F}_{N,y}(x) := \frac{1}{N} \sum_{j=1}^N \mathbf{1}_{\{Z_j \leq x\}}.$$

Almost surely as $N \rightarrow \infty$,

$$(\bar{Z}, \hat{v}, \hat{\gamma}_3, \hat{\gamma}_4) \longrightarrow \left(0, 1, 0, 3 - \frac{3}{8\sigma_y^4} \sum_{p \leq y} p^{-2}\right), \quad \sup_{x \in \mathbb{R}} |\widehat{F}_{N,y}(x) - F_y(x)| \longrightarrow 0.$$

Proof. Apply the strong law and Glivenko–Cantelli [3, 4, 6].

$$Z_j \stackrel{d}{=} \frac{S_y(0)}{\sigma_y} \xrightarrow{2.1.5} Z_j \stackrel{d}{=} \frac{S_y(t)}{\sigma_y}, \quad |Z_j| \leq \sigma_y^{-1} \sum_{p \leq y} p^{-1/2} < \infty.$$

$$(Z_j)_{j \geq 1} \text{ i.i.d.} \implies \frac{1}{N} \sum_{j=1}^N Z_j^r \longrightarrow \mathbb{E}(Z_1^r) \quad (1 \leq r \leq 4) \text{ a.s.}$$

$$\frac{1}{N} \sum_{j=1}^N Z_j^r \longrightarrow \mathbb{E}(Z_1^r) \quad (1 \leq r \leq 4) \xrightarrow{2.2.1, 3.2.1} (\bar{Z}, \hat{v}, \hat{\gamma}_3, \hat{\gamma}_4) \longrightarrow \left(0, 1, 0, 3 - \frac{3}{8\sigma_y^4} \sum_{p \leq y} p^{-2}\right).$$

$$\widehat{F}_{N,y}(x) = \frac{1}{N} \sum_{j=1}^N \mathbf{1}_{\{Z_j \leq x\}} \implies \sup_{x \in \mathbb{R}} |\widehat{F}_{N,y}(x) - F_y(x)| \longrightarrow 0 \quad \text{a.s.}$$

This proves both almost-sure limits. □

Example 3.2.4 (Experiment A). Write

$$\pi(y) := \#\{p \leq y\}, \quad \gamma_4(y) := 3 - \frac{3}{8\sigma_y^4} \sum_{p \leq y} p^{-2}.$$

Definition 3.2.2 gives

y	$\pi(y)$	σ_y^2	$\gamma_4(y)$	$(\bar{Z}, \hat{v}, \hat{\gamma}_3, \hat{\gamma}_4)$
10^3	168	1.0990	2.8596	$(-0.0032, 0.9965, 0.0197, 2.8397)$
10^4	1229	1.2415	2.8900	$(-0.0035, 1.0178, 0.0143, 2.9214)$
10^5	9592	1.3526	2.9073	$(0.0099, 1.0189, -0.0057, 2.9316)$

Figure 5 records $y = 10^3$.

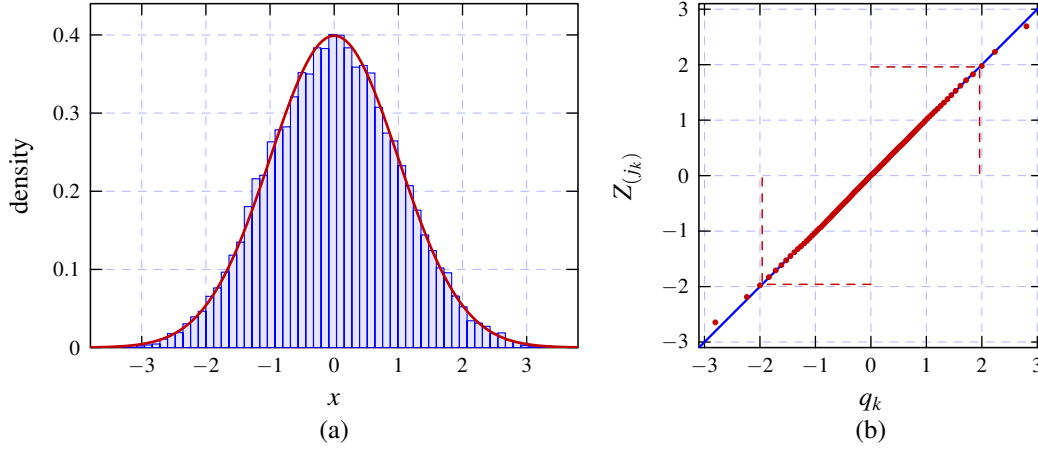


FIGURE 5. For $y = 10^3$ and $N = 2 \cdot 10^4$, the histogram and quantile plot nearly match the normal density and identity line; the inward tail deviation is consistent with Proposition 3.2.1.

3.3. Arithmetic scope. The surrogate retains Euler weights and removes arithmetic phase coupling [10, Chapter 1], [1, Chapter 1], [2, Part III], [12].

Remark 3.3.1 (What the surrogate retains).

$$\log(\zeta(s)) = \sum_p \sum_{k \geq 1} \frac{p^{-ks}}{k} \quad (\Re(s) > 1), \quad p^{-it} = e^{-it \log(p)} \quad [10, \text{Chapter 1}], [1, \text{Chapter 1}].$$

$$\log(Z_y(t)) = \sum_{p \leq y} \sum_{k \geq 1} \frac{\xi_p^k p^{-ikt}}{k p^{k/2}}, \quad \xi_p p^{-it} = e^{i(\theta_p - t \log(p))}, \quad (\theta_p)_p \text{ independent.}$$

4. PROOF OF THEOREM A, THEOREM B

Theorem 4.0.1 (Stationarity, variance, and Gaussian limit)). *The assertions of Theorem A hold.*

(1) For every $m \in \mathbb{N} \setminus \{0\}$, $h, t_1, \dots, t_m \in \mathbb{R}$, and $y \geq 2$,

$$\mathcal{L}\left((\mathbf{Y}_y(t_j + h))_{j=1}^m\right) = \mathcal{L}\left((\mathbf{Y}_y(t_j))_{j=1}^m\right).$$

Proof. Set

$$\theta'_p := (\theta_p - h \log p) \bmod 2\pi.$$

Lemma 2.1.3 and independence give

$$(\theta'_p)_{p \leq y} \stackrel{d}{=} (\theta_p)_{p \leq y}, \quad (U_p(t_j + h))_{\substack{p \leq y \\ 1 \leq j \leq m}} = ((\theta'_p - t_j \log p) \bmod 2\pi)_{\substack{p \leq y \\ 1 \leq j \leq m}} \stackrel{d}{=} (U_p(t_j))_{\substack{p \leq y \\ 1 \leq j \leq m}},$$

which proves the asserted joint law. $\square$

(2) For every $t \in \mathbb{R}$,

$$\sigma_y^2 = \frac{1}{2} \log \log y + \frac{B_1}{2} + O\left(\frac{1}{\log y}\right),$$

$$\text{Var}(X_y(t)) = \frac{1}{2} \log \log y + \frac{B_1}{2} + C_R + O\left(\frac{1}{\log y}\right).$$

Proof. Proposition 3.1.1 gives

$$\begin{aligned} \text{Var}(X_y(t)) &= \sigma_y^2 + \frac{1}{2} \sum_{p \leq y} \sum_{k \geq 2} \frac{1}{k^2 p^k} = \sigma_y^2 + C_R + O(y^{-1}) \implies \sigma_y^2 \xrightarrow{2.2.2} \frac{1}{2} \log \log y + \frac{B_1}{2} + O\left(\frac{1}{\log y}\right) \\ &\implies \text{Var}(X_y(t)) = \frac{1}{2} \log \log y + \frac{B_1}{2} + C_R + O\left(\frac{1}{\log y}\right). \end{aligned}$$

□

(3) For every $t \in \mathbb{R}$,

$$\frac{X_y(t)}{\sigma_y} \Rightarrow \mathcal{N}(0, 1) \quad (y \rightarrow \infty).$$

Proof.

$$\frac{S_y(t)}{\sigma_y} \Rightarrow \mathcal{N}(0, 1), \quad \frac{R_y(t)}{\sigma_y} \xrightarrow{\mathbb{P}} 0 \xrightarrow{2.3.5, 2.2.7} \frac{X_y(t)}{\sigma_y} = \frac{S_y(t) + R_y(t)}{\sigma_y} \Rightarrow \mathcal{N}(0, 1).$$

□

Theorem 4.0.2 (Quantitative Gaussian approximation). *Define*

$$D_y^S(t) := \sup_{x \in \mathbb{R}} \left| \mathbb{P}\left(\frac{S_y(t)}{\sigma_y} \leq x\right) - \Phi(x) \right|, \quad D_y^X(t) := \sup_{x \in \mathbb{R}} \left| \mathbb{P}\left(\frac{X_y(t)}{\sigma_y} \leq x\right) - \Phi(x) \right|.$$

Then the assertions of Theorem B hold uniformly in t .

(1) For every $y \geq 2$,

$$D_y^S(t) \leq \frac{4C_{\text{BE}}}{3\pi\sigma_y^3} \sum_{p \leq y} p^{-3/2} < \frac{C_{\text{BE}}}{2\sigma_y^3}.$$

Proof. Theorem 3.1.3 and Lemma 3.1.4 give

$$D_y^S(t) \leq \frac{C_{\text{BE}}}{\sigma_y^3} \sum_{p \leq y} \mathbb{E}|X_{p,y}(t)|^3 = \frac{4C_{\text{BE}}}{3\pi\sigma_y^3} \sum_{p \leq y} p^{-3/2} \implies \sum_{p \leq y} p^{-3/2} < P(3/2), \quad \frac{4P(3/2)}{3\pi} < \frac{1}{2},$$

the result follows. □

(2) If $\sigma_y \geq 1$, then

$$D_y^X(t) \leq \left(\frac{1}{\sqrt{2\pi}} + \frac{1}{6} + \frac{C_{\text{BE}}}{2} \right) \sigma_y^{-2/3}.$$

Proof. For $\delta > 0$, set

$$q_\delta := \mathbb{P}(|R_y(t)| > \delta \sigma_y).$$

Since $X_y = S_y + R_y$,

$$\mathbb{P}\left(\frac{S_y(t)}{\sigma_y} \leq x - \delta\right) - q_\delta \leq \mathbb{P}\left(\frac{X_y(t)}{\sigma_y} \leq x\right) \leq \mathbb{P}\left(\frac{S_y(t)}{\sigma_y} \leq x + \delta\right) + q_\delta.$$

Proposition 2.2.6, Chebyshev's inequality, and Lemma A.1.3 yield

$$q_\delta < \frac{1}{6\delta^2\sigma_y^2}, \quad D_y^X(t) \leq D_y^S(t) + \frac{\delta}{\sqrt{2\pi}} + \frac{1}{6\delta^2\sigma_y^2}.$$

Choose $\delta = \sigma_y^{-2/3}$. Since $\sigma_y \geq 1$,

$$D_y^X(t) < \frac{C_{\text{BE}}}{2\sigma_y^3} + \left(\frac{1}{\sqrt{2\pi}} + \frac{1}{6} \right) \sigma_y^{-2/3} \implies D_y^X(t) \leq \left(\frac{1}{\sqrt{2\pi}} + \frac{1}{6} + \frac{C_{\text{BE}}}{2} \right) \sigma_y^{-2/3}.$$

□

Part 2. Appendix

APPENDIX A. FOUNDATIONAL TOOLS

A.1. Moments and empirical limits.

Fact A.1.1 (Independent-sum moments, [5, 6]). *Let $Y_1, \dots, Y_n \in L^4$ be independent and centered, and set $T := \sum_{j=1}^n Y_j$, $v^2 := \sum_{j=1}^n \mathbb{E}(Y_j^2)$. Then*

$$\mathbb{E}(T^3) = \sum_{j=1}^n \mathbb{E}(Y_j^3), \quad \mathbb{E}(T^4) = 3v^4 + \sum_{j=1}^n (\mathbb{E}(Y_j^4) - 3\mathbb{E}(Y_j^2)^2).$$

Fact A.1.2 (Strong law and Glivenko–Cantelli, [3, 4, 6]). *Let $Y_1, Y_2, \dots$ be i.i.d., $Y_1 \in L^4$, and put*

$$F(x) := \mathbb{P}(Y_1 \leq x), \quad F_N(x) := \frac{1}{N} \sum_{j=1}^N \mathbf{1}_{\{Y_j \leq x\}}.$$

Almost surely,

$$\frac{1}{N} \sum_{j=1}^N Y_j^r \longrightarrow \mathbb{E}(Y_1^r) \quad (1 \leq r \leq 4), \quad \sup_{x \in \mathbb{R}} |F_N(x) - F(x)| \longrightarrow 0.$$

Lemma A.1.3 (Lipschitz bound for the normal distribution function). *For $x \in \mathbb{R}$ and $\delta > 0$,*

$$|\Phi(x \pm \delta) - \Phi(x)| = \left| \int_x^{x \pm \delta} \varphi(u) du \right| \leq \delta \sup_{u \in \mathbb{R}} \varphi(u) = \frac{\delta}{\sqrt{2\pi}}.$$

Fact A.1.4 (Lévy’s continuity theorem, [3]). *For real random variables Y_n and Y ,*

$$Y_n \Rightarrow Y \iff \mathbb{E}[e^{i\lambda Y_n}] \longrightarrow \mathbb{E}[e^{i\lambda Y}] \quad (\lambda \in \mathbb{R}).$$

Proof. [3, §26], [4, §3.3], [5, Chapter XV], [8, Chapter 8]. □

Fact A.1.5 (Chebyshev’s inequality, [6]). *For a random variable X with $\mathbb{E}[X^2] < \infty$ and every $\varepsilon > 0$,*

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq \varepsilon) \leq \frac{\text{Var}(X)}{\varepsilon^2}.$$

Proof. [6, Chapter 3]. □

Fact A.1.6 (Slutsky’s theorem, [3]). *For real random variables Y_n, Y, W_n on a common probability space,*

$$Y_n \Rightarrow Y, \quad W_n \xrightarrow{\mathbb{P}} 0 \implies Y_n + W_n \Rightarrow Y.$$

Proof. [3, §25], [4, §3.2], [6, Chapter 5]. □

APPENDIX B. SOURCE CODE

B.1. Experiment A.

LISTING 1. `experiment_a.py`: the moments of Example 3.2.4 and the coordinates of Figure 5.

```

1 import math
2 from statistics import NormalDist
3 import numpy as np
4
5 N, SEED, BATCH = 20_000, 7, 256
6
7 def primes(y):
8     keep = np.ones(y + 1, dtype=bool)
```

```

9     keep[:2] = False
10    for p in range(2, math.isqrt(y) + 1):
11        if keep[p]:
12            keep[p * p::p] = False
13    return np.flatnonzero(keep).astype(float)
14
15    def sample(y):
16        p = primes(y)
17        root = np.sqrt(p)
18        sigma2 = .5 * np.sum(p ** -1)
19        rng = np.random.default_rng(SEED)
20        z = np.empty(N)
21        for lo in range(0, N, BATCH):
22            hi = min(lo + BATCH, N)
23            theta = rng.uniform(0, 2 * np.pi, (hi - lo, p.size))
24            np.cos(theta, out=theta)
25            np.divide(theta, root, out=theta)
26            z[lo:hi] = theta.sum(axis=1) / np.sqrt(sigma2)
27        return z, p, sigma2
28
29    def moments(z):
30        mean = z.mean()
31        centered = z - mean
32        variance = np.mean(centered ** 2)
33        standardized = centered / np.sqrt(variance)
34        return mean, variance, np.mean(standardized ** 3), np.mean(standardized ** 4)
35
36    def write(name, x, y, pattern):
37        with open(name, "w", encoding="utf-8") as output:
38            output.write(" ".join(pattern % pair for pair in zip(x, y)) + "\n")
39
40    for y in (10 ** 3, 10 ** 4, 10 ** 5):
41        z, p, sigma2 = sample(y)
42        mean, variance, skew, kurtosis = moments(z)
43        exact = 3 - 3 * np.sum(p ** -2) / (8 * sigma2 ** 2)
44        print(y, p.size, *(f"{v:.4f}" for v in
45            (sigma2, exact, mean, variance, skew, kurtosis)))
46        if y == 10 ** 3:
47            density, edge = np.histogram(z, bins=60, density=True)
48            write("hist_le3.txt", edge,
49                np.r_[density, density[-1]], "(%.3f,%.4f)")
50            order = np.sort(z)
51            u = np.linspace(.0025, .9975, 99)
52            index = np.floor(u * (N - 1) + .5).astype(int)
53            normal = NormalDist()
54            q = [normal.inv_cdf((j + .5) / N) for j in index]
55            write("qq_le3.txt", q, order[index], "(%.3f,%.3f)")

```

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